TruERP & ChatGPT Setup Guide

MCP Server, OAuth 2.0 & Custom GPT Actions Integration

PRODUCTION READY

1. System Architecture

The solution connects OpenAI's ChatGPT directly to your TruERP system using OAuth 2.0 Authorization Code flow and REST/SSE APIs.

> **Configured Public Tunnel Domain**
>
> `https://operation-common-joseph-univ.trycloudflare.com`

- **TruERP Go Backend (Port 8088):** Handles core billing, inventory, HR, accounting, and hosts the OAuth 2.0 authorization server.
- **TruERP MCP Server (Port 8089):** Exposes 526 tools via HTTP/SSE and generates the OpenAPI 3.1 schema for ChatGPT.
- **Cloudflare Tunnel:** Exposes port 8089 to public HTTPS for secure OpenAI communication.

2. Starting the Services

Step 2.1: Start TruERP Backend

```
cd backend
go run .
```

Runs on `http://localhost:8088`. Initializes database schema and default OAuth client (`truerp-mcp-client` / `truerp-mcp-secret`).

Step 2.2: Start TruERP MCP Server in HTTP/SSE Mode

```
cd mcp-server
npm install
npm run build
node dist/index.js --transport sse --port 8089 --host 0.0.0.0
```

Step 2.3: Expose via Cloudflare Tunnel

```
cloudflared tunnel --url http://localhost:8089
```

3. Configuring ChatGPT Custom GPT (Step-by-Step)

Step 3.1: Create a Custom GPT

1. Navigate to chatgpt.com → **My GPTs** → **+ Create a GPT**.
2. Click the **Configure** tab and set:
   - **Name:** TruERP Business Assistant
   - **Description:** AI ERP assistant for invoices, stock, customers, and reports.

Step 3.2: Import OpenAPI Action Schema

Under **Actions**, click **Create new action**. In the **Schema** box, paste:

```
https://operation-common-joseph-univ.trycloudflare.com/openapi.json
```

Or paste the contents of `mcp-server/chatgpt-actions-openapi.json`.

Step 3.3: Configure OAuth 2.0 Authentication

Under **Authentication**, select **OAuth** and enter:

Setting	Value
Authentication Type	`OAuth`
Client ID	`truerp-mcp-client`
Client Secret	`truerp-mcp-secret`
Authorization URL	`https://operation-common-joseph-univ.trycloudflare.com/oauth/authorize`
Token URL	`https://operation-common-joseph-univ.trycloudflare.com/oauth/token`
Scope	`read write admin erp`
Token Exchange Method	`Default (POST request)`

Step 3.4: ChatGPT Callback URL

ChatGPT displays a generated callback URL like `https://chatgpt.com/aip/g-.../oauth/callback`. The default client `truerp-mcp-client` is pre-configured to accept it automatically.

4. Testing the Integration

1. In the Custom GPT preview, ask: *"Show our dashboard overview and sales stats."*
2. Click **Log In** when prompted by ChatGPT.

3. Enter your TruERP email and password in the authorization screen and click **Authorize Access**.
4. ChatGPT receives the token, calls your live API, and returns real-time data!

5. Public Endpoints Summary

Endpoint	Description
`GET /openapi.json`	OpenAPI 3.1 action specification for ChatGPT
`GET /oauth/authorize`	Interactive user login and consent UI
`POST /oauth/token`	OAuth 2.0 token issuance & refresh
`GET /.well-known/oauth-authorization-server`	RFC 8414 OAuth 2.0 metadata
`GET /sse`	MCP SSE connection for MCP-compatible AI agents
`GET /health`	Health check